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**CO3053 ASSIGNMENT 2: DESIGN AND IMPLEMENTATION OF A
WASHING MACHINE CONTROL UNIT REPORT**

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1. INTRODUCTION

1.1 Purpose and scope

This report presents the design and implementation of the control unit of a coin-operated washing machine, as required by Assignment 2 of course CO3053. The control unit reads three push buttons (STOP, RUN and PAUSE) and a coin acceptor, and drives two indicator LEDs, a red LED (RLED) and a blue LED (BLED), together with the washing motor.

The report analyses the requirements, models the behavior as a finite state machine, describes the logic with flowcharts, presents the C implementation and reports the results of a simulated test campaign.

2. PROBLEM ANALYSIS

2.1 Requirements

Table 1 lists the requirements extracted from the assignment. Each requirement has an identifier that is used later to trace the tests.

ID	Requirement
R1	The machine is controlled by three buttons: STOP, RUN and PAUSE.
R2	The machine has two LEDs: a red LED (RLED) and a blue LED (BLED).
R3	RLED is on when the machine is on standby, i.e., available to serve.
R4	RLED is blinking if the machine has errors.
R5	BLED is on when the machine is ready to execute.
R6	BLED is blinking when the machine is running (doing washing).
R7	The machine can execute only once at least 50 cents has been added.
R8	The machine accepts only 10-cent, 20-cent and 50-cent coins.
R9	When there is enough money (BLED is on), pressing RUN starts the machine. A 30-minute timer is activated and the money is cleared without returning the surplus.

R10	While the machine is running, pressing PAUSE pauses it, but the timer keeps counting down. Pressing RUN again re-executes the machine.
R11	A running machine terminates automatically after 30 minutes (washing cycle) or when STOP is pressed twice (forced stop).

Table 1. Requirements of the washing machine control unit.

2.2 Interpretation and assumptions

The assignment leaves some behaviors open. The following decisions were taken so that the design is complete and can be tested:

- Each state lights the LEDs exclusively: RLED is on only on standby, and in the ready state BLED is on while RLED is off.
- A paused machine keeps BLED steady on, because it is ready to resume when RUN is pressed. The motor is stopped during the pause.
- The assignment does not define the error conditions. An error is defined as a coin other than 10, 20 or 50 cents inserted while the machine is waiting for money. The error is cleared by pressing STOP once, which also clears the credit.
- "STOP pressed twice" means two presses within 3 seconds. The first press only opens the 3-second window, so a single accidental press does not interrupt the washing.
- Money above 50 cents is accepted while the machine is ready, and it is cleared without change when RUN is pressed, as required by R9.
- Coins inserted while the machine is running or paused are rejected and are not added to the credit.
- RUN pressed with less than 50 cents, and PAUSE or STOP pressed in a state where they have no meaning, are ignored.
- When a cycle ends, by timeout or by forced stop, the credit is zero and the machine returns to standby with RLED on.
- The timer keeps counting during a pause. If it reaches zero while the machine is paused, the machine terminates.

3. SYSTEM DESIGN

3.1 Inputs and outputs

Table 2 describes the signals of the control unit.

Signal	Type	Description
RUN button	Input	Starts the machine when it is ready, and resumes it when it is paused.
PAUSE button	Input	Pauses the machine while it is running.
STOP button	Input	Pressed twice to force a running or paused machine to stop. Pressed once to clear an error.
Coin acceptor	Input	Reports the value of an inserted coin in cents. Valid values are 10, 20 and 50.
RLED (red)	Output	On: standby. Blinking: error.
BLED (blue)	Output	On: ready or paused. Blinking: running.
Motor enable	Output	On only while the machine is running.

Table 2. Inputs and outputs.

3.2 Controller variables

- **state**: the current state (IDLE, READY, RUNNING, PAUSED or ERROR).
- **credit**: the money inserted so far, in cents (0 to 9999).
- **remain**: the number of 10 ms ticks left in the washing cycle.
- **stop_cnt** and **stop_win**: the number of STOP presses in the current window and the ticks left in that window.
- **blink_cnt** and **blink_on**: the tick counter and the phase used to blink an LED at 1 Hz.
- **rled**, **bled** and **motor**: the output levels computed from the state.

3.3 States and outputs

The machine has five states. The outputs depend on the state only, plus the blinking phase, as shown in Table 4.

State	Meaning	RLED	BLED	Motor
IDLE	Standby, credit below 50 cents	On	Off	Off

READY	Credit of at least 50 cents, waiting for RUN	Off	On	Off
RUNNING	Washing, timer counting down	Off	Blinking	On
PAUSED	Washing suspended, timer still counting down	Off	On	Off
ERROR	Invalid coin detected	Blinking	Off	Off

Table 4. States and their outputs.

3.4 State machine

Figure 1 shows the state machine. At power-on the machine enters IDLE, which is marked by the filled circle. Table 5 gives the meaning of each numbered transition.

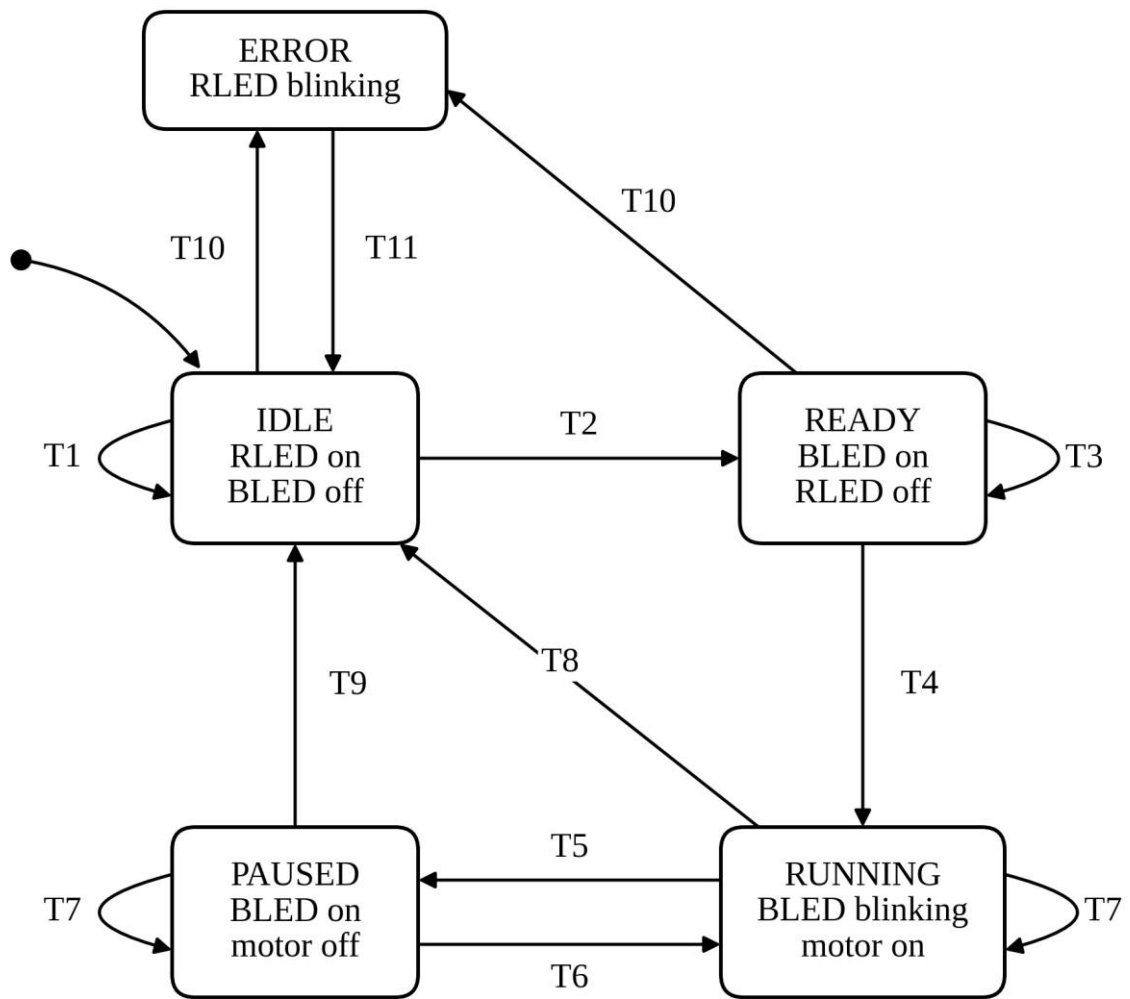


Figure 1. State machine of the washing machine control unit.

No.	From	To	Event and condition	Action
T1	IDLE	IDLE	Valid COIN, credit after adding is below 50	credit = credit + coin
T2	IDLE	READY	Valid COIN, credit after adding is at least 50	credit = credit + coin
T3	READY	READY	Valid COIN	credit = credit + coin
T4	READY	RUNNING	RUN	credit = 0, remain = 180000

T5	RUNNING	PAUSED	PAUSE	None
T6	PAUSED	RUNNING	RUN	None
T7	RUNNING or PAUSED	Same state	STOP, first press	Open the 3 s window
T8	RUNNING	IDLE	STOP, second press within the window, or remain = 0	remain = 0, credit = 0
T9	PAUSED	IDLE	STOP, second press within the window, or remain = 0	remain = 0, credit = 0
T10	IDLE or READY	ERROR	COIN with a value other than 10, 20 or 50	None
T11	ERROR	IDLE	STOP	credit = 0

Table 5. State transitions.

3.5 Flowcharts

The behavior is described by five flowcharts. Figure 2 shows the main loop, which runs once per 10 ms tick. The loop waits for the tick, samples and debounces the buttons, reads the coin acceptor, converts each input into an event for the controller, lets the controller advance its timers and finally drives the LEDs and the motor.

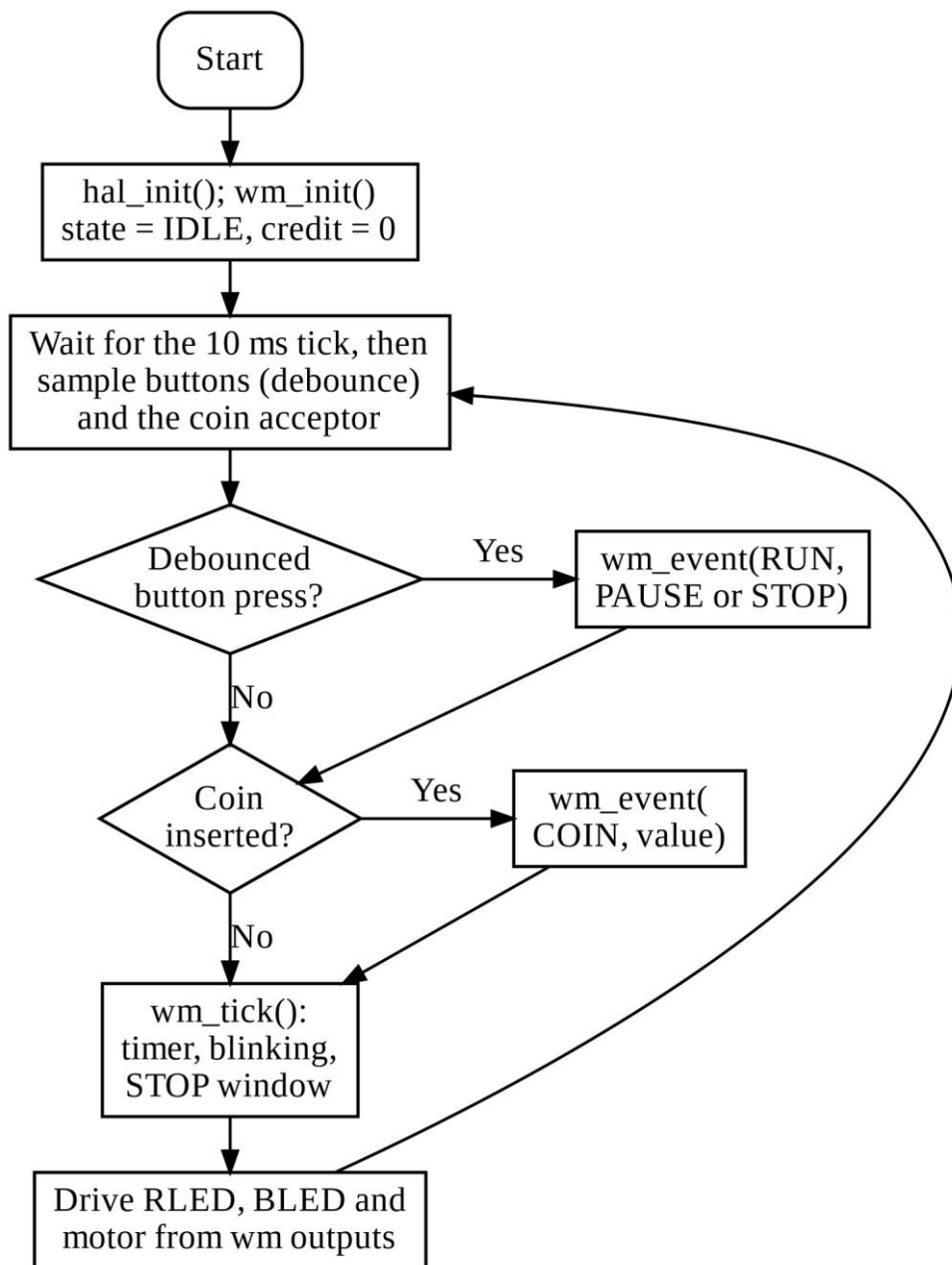


Figure 2. Flowchart of the main loop.

Figure 3 shows the handling of an event when the machine is in IDLE or READY. An invalid coin leads to ERROR. RUN in READY starts the cycle, clears the credit and loads the 30-minute timer. A valid coin is added to the credit, and IDLE becomes READY as soon as the credit reaches 50 cents.

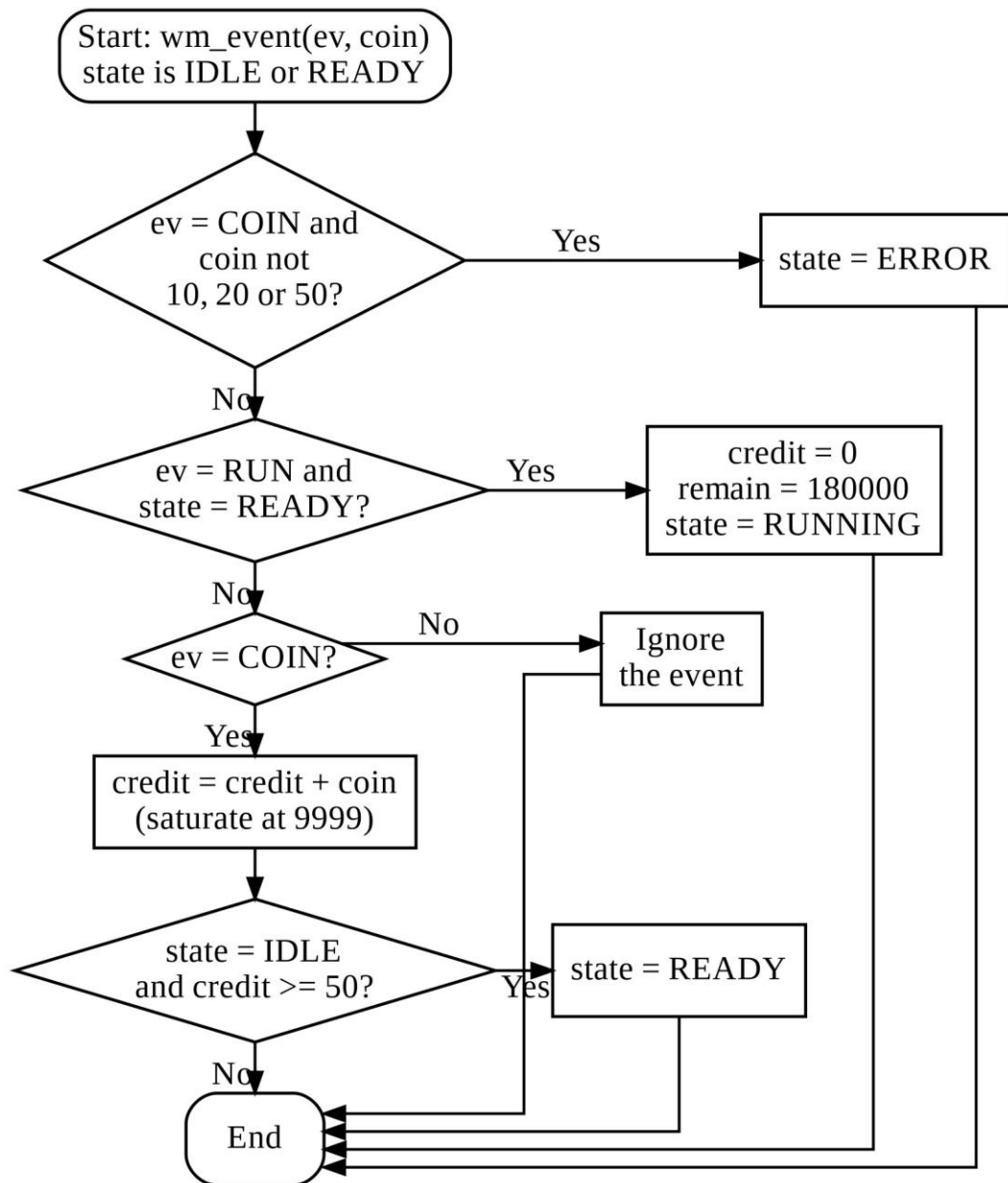


Figure 3. Flowchart of event handling in IDLE and READY.

Figure 4 shows the handling of an event in RUNNING, PAUSED and ERROR. STOP clears an error, PAUSE suspends a running machine, RUN resumes a paused machine, and STOP in RUNNING or PAUSED is passed to the double-press logic of Figure 5. All other events are ignored.

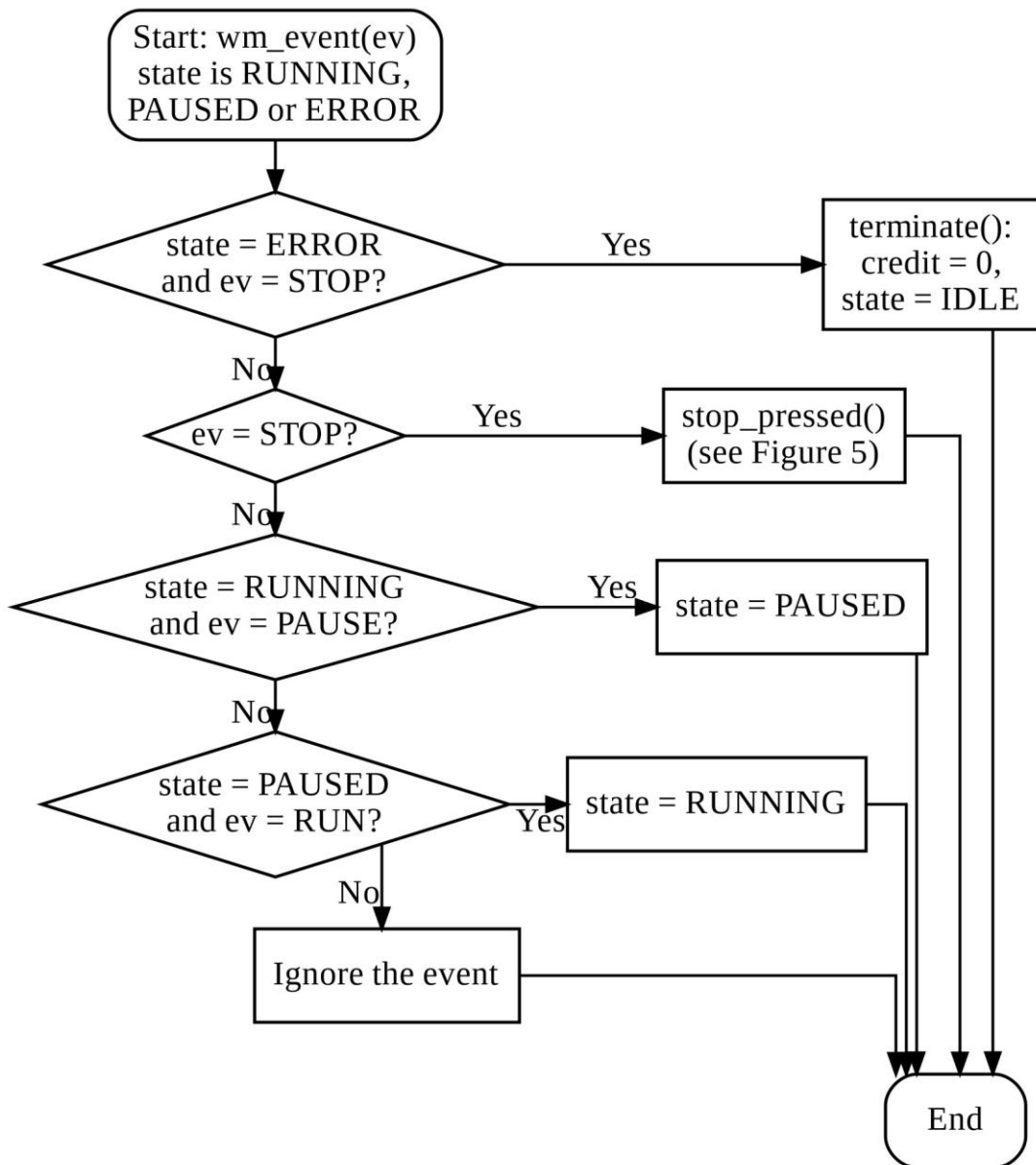


Figure 4. Flowchart of event handling in RUNNING, PAUSED and ERROR.

Figure 5 shows the double-press logic. The first press increments the counter and opens a 3-second window. The second press inside the window terminates the cycle.

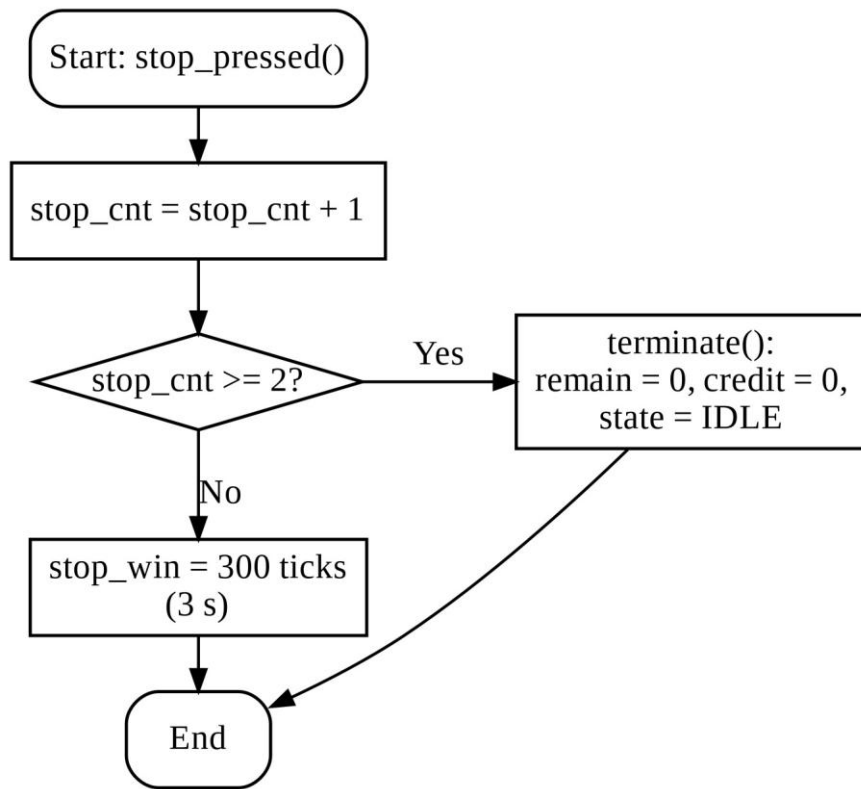


Figure 5. Flowchart of the STOP double-press logic.

Figure 6 shows the periodic update. While the machine is running or paused, the cycle timer is decremented and the machine terminates when it reaches zero. The STOP window is counted down, and the blinking phase is toggled every 50 ticks in RUNNING and ERROR.

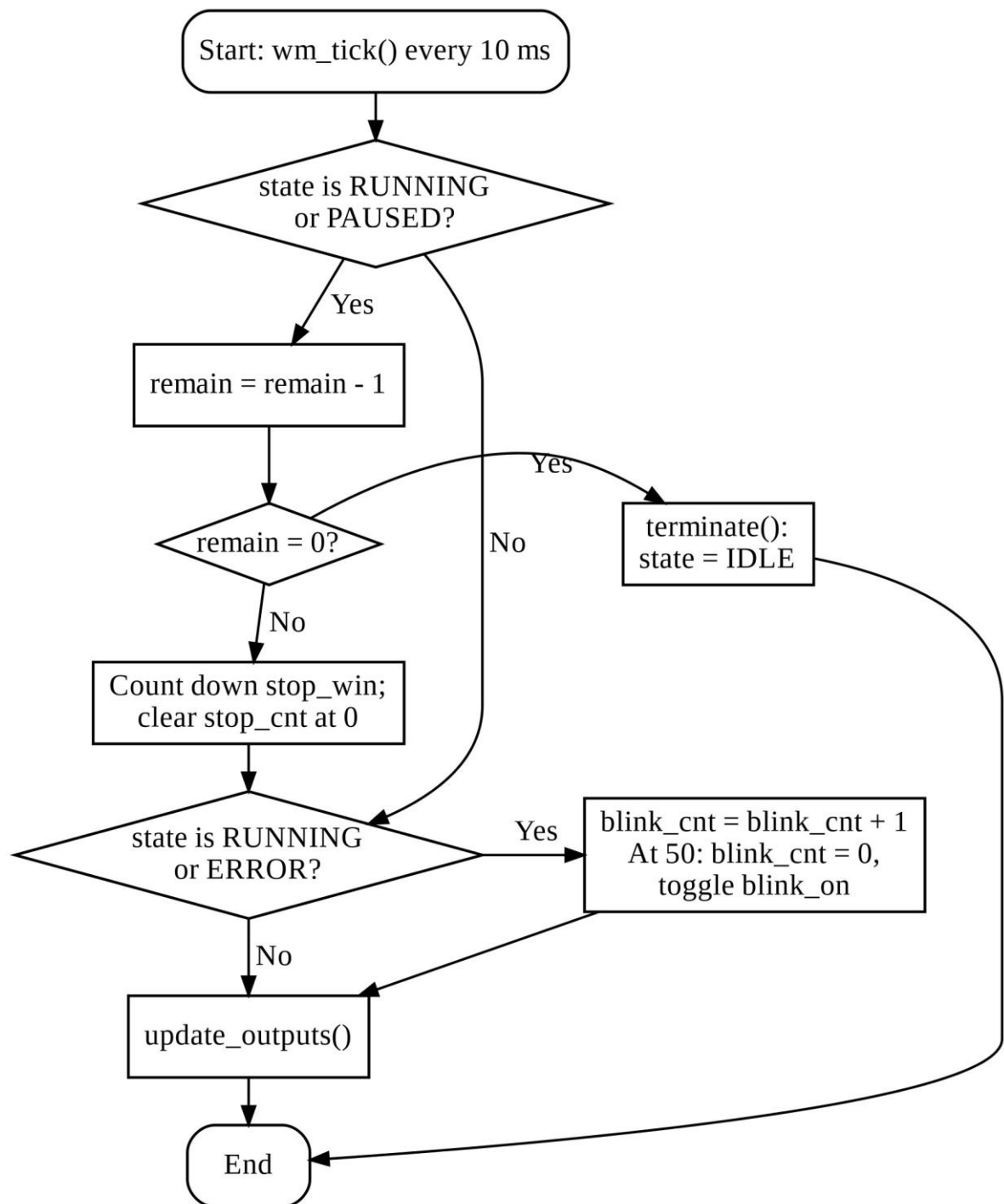


Figure 6. Flowchart of the 10 ms periodic update.

4. IMPLEMENTATION

4.1 Software architecture

The software is divided into three layers so that the control logic does not depend on the hardware. Table 6 lists the source files.

File	Role
hal.h	Declares the functions that the target board must provide: reading the buttons and the coin acceptor, driving the LEDs and the motor, and waiting for the 10 ms tick.
main.c	Application loop. Waits for the tick, debounces the buttons, turns the inputs into events, calls the controller and drives the outputs.
wm.h, wm.c	Control logic. The finite state machine, written in portable C with no hardware access.
test_wm.c	Host-side test program with 16 test cases (Section 5).

Table 6. Source files.

4.2 Control logic

The controller is event driven. All of its data is kept in one structure of type `wm_t`, and it offers three public functions:

- **`wm_init()`** puts the machine in IDLE with no credit.
- **`wm_event()`** handles one RUN, PAUSE, STOP or COIN event according to the current state (Figures 3 and 4).
- **`wm_tick()`** is called every 10 ms. It counts down the cycle timer and the STOP window, toggles the blinking phase and refreshes the outputs (Figure 6).

Three private helper functions keep the code simple. The function `enter()` changes the state, clears the per-state counters and refreshes the outputs, so every state starts from a known condition. The function `terminate()` ends a cycle or clears an error by zeroing remain and credit and returning to IDLE. The function `stop_pressed()` implements the double-press logic (Figure 5). The outputs are computed in one place, `update_outputs()`, directly from the state, which makes the LED table of Table 4 easy to check against the code.

4.3 Timing and button debouncing

The washing time is measured by counting ticks of the hardware timer instead of reading a software clock, so it does not drift. The counter remain is decremented in both RUNNING and PAUSED, which implements the requirement that the timer keeps

counting down during a pause. A pause therefore does not extend the cycle, and the cycle can end while the machine is paused.

A button is accepted as pressed when its raw level differs from the stable level for three consecutive ticks (30 ms). Only the press edge produces an event, so holding a button down generates one event and not a stream of events. This is important for STOP, because a long press must not be counted as two presses.

4.4 Porting to a target board

To run the control unit on hardware, only the functions declared in `hal.h` must be implemented. The file `wm.c` does not need to change. The steps are:

- configure a hardware timer that raises a flag every 10 ms and implement `hal_wait_tick_10ms()`;
- configure the GPIO pins for the three buttons, the two LEDs and the motor driver, and implement the read and write functions;
- connect the coin acceptor, and implement `hal_coin_poll()` so that it returns the coin value in cents once per inserted coin, and 0 otherwise.

5. TESTING AND VERIFICATION

5.1 Method

The control logic was verified on a PC. The file `wm.c` was compiled with `gcc` using the options `-Wall -Wextra -pedantic -std=c99`, without warnings, together with the test program `test_wm.c`. The test program calls `wm_event()` and `wm_tick()` directly and simulates time by calling `wm_tick()` the required number of times: 100 calls represent one second and 180000 calls represent the complete 30-minute cycle. Each test case checks the state, the credit, the timer and the outputs.

6. CONCLUSION

This report presented the design and implementation of the control unit of a coin-operated washing machine. The behavior was modeled as a five-state machine (IDLE, READY, RUNNING, PAUSED and ERROR) with eleven transitions, described with one state diagram and five flowcharts, and implemented in portable C. The 30-minute cycle is measured by counting 10 ms ticks and keeps counting during a pause. The double-press logic of the STOP button and the clearing of the credit without change follow the assignment. The simulated tests show that requirements R3 to R11 are met, and requirements R1 and R2 are satisfied by the input and output design.

Possible extensions are:

- a display of the credit and of the remaining washing time;
- the return of the surplus money, which the assignment explicitly excludes;
- several error conditions, for example a door or water-level sensor, shown with the same blinking RLED;
- a watchdog timer to restart the controller if the main loop stops running.

Source_code: https://github.com/amnt123/CO3053_Assignment2